

Gegeben: $v = 0,48 \frac{m}{s}$; $\omega = 1,65 \frac{m}{s^2}$

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am 24.7.20

11 (6.41)

Daten:

$l = 0,3 m$

$d = 0,1 m$

$T = ?$

$L = ?$

Zeichnung:

Gegeben $A = d$, $B = l - d = 0,2 m$

Wegen L :

$L = \frac{A^2 + B^2}{2} = \frac{0,1^2 + 0,2^2}{2} =$

$= 0,025 m = 0,25 m$



$T = 2\pi \sqrt{\frac{L}{g}} = 2\pi \sqrt{\frac{0,25}{9,80}} \approx 3,17 s$

Gegeben: $L = 0,25 m$, $T = 3,17 s$

Daten:

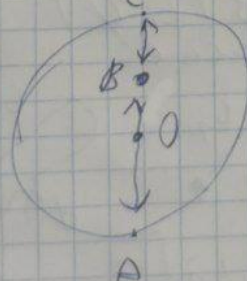
$R = 0,24 m$

$\frac{AB}{BC} = \frac{1}{3}$

$BC = 3$

$L, T = ?$

Zeichnung:



$AB = \frac{3}{4} \cdot 0,24 = 0,18 m$

$BC = \frac{1}{4} \cdot 0,24 = 0,06 m$

$AO = \frac{3}{4} \cdot 0,24 = 0,18 m$

$BO = 0,18 - 0,06 = 0,12 m$

11 (6.46)

$$L = \frac{J}{m(0.5)} = \frac{\frac{1}{2} m R^2}{m(0.5)} = \frac{R^2}{2 \cdot 0.5} = \frac{0.24^2}{2 \cdot 0.06} = 0.48 \text{ m}$$

$$T = 2\pi \sqrt{\frac{L}{g}} = 2\pi \sqrt{\frac{0.48}{9.81}} \approx 1.39 \text{ s}$$

Answer: $L = 0.48 \text{ m}$, $T \approx 1.39 \text{ s}$

$J_0 = \frac{1}{2} m R^2$, no πR . Wrong!

$$J = J_0 + m \left(\frac{R}{2}\right)^2 = \frac{1}{2} m R^2 + \frac{1}{4} m R^2 = \frac{3}{4} m R^2$$

$$L = \frac{J}{m g d} = \frac{\frac{3}{4} m R^2}{m g d} = \frac{3 R^2}{4 g d} = \frac{3 \cdot 0.24^2}{4 \cdot 9.81 \cdot 0.12} \approx 0.37 \text{ m}$$

$$T = 2\pi \sqrt{\frac{L}{g}} = 2\pi \sqrt{\frac{0.37}{9.81}} \approx 0.386 \text{ s}$$

Answer: $L \approx 0.37 \text{ m}$, $T \approx 0.386 \text{ s}$

3 (6.4)

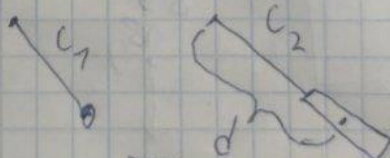
Given:

$$L_1 = 40 \text{ cm} = 0.4 \text{ m}$$

$$L_2 = 0.6 \text{ m}$$

d = ?

Diagram:



$$T_1 = 2\pi \sqrt{\frac{L_1}{g}} = 2\pi \sqrt{\frac{0.4}{9.81}} \approx 1.26 \text{ s}$$

$$T_2 = 2\pi \sqrt{\frac{J}{m g d}}$$

#XAJABAPM

23 am 24.10
13 (~6.41)

$$J_{\text{Kugel}} = \frac{1}{3} m l_2^2$$

10 m. Kugelmoment

$$J = J_{\text{Kugel}} + m d^2 = \frac{1}{12} m l_2^2 + m d^2$$

$$\Rightarrow T_2 = 2\pi \sqrt{\frac{\frac{1}{12} m l_2^2 + m d^2}{m g d}} = 2\pi \sqrt{\frac{\frac{1}{12} l_2^2 + d^2}{g d}}$$

System:

$$T_1 = T_2$$

$$2\pi \sqrt{\frac{\frac{1}{12} l_2^2 + d^2}{g d}} = 2\pi \sqrt{\frac{l_1}{g}}$$

$$\frac{\frac{1}{12} l_2^2 + d^2}{g d} = \frac{l_1}{g} \quad \frac{1}{12} \frac{l_2^2}{d} + d = l_1$$

$$\frac{l_2^2}{12d} + d = l_1 \quad | \cdot 12d$$

$$12d^2 - 12d l_1 + l_2^2 = 0$$

$$12d^2 - 4,8d + 0,36 = 0 \quad | :12$$

$$d^2 - 0,4d + 0,03 = 0$$

$$\Rightarrow d_1 = 0,3, d_2 = 0,1$$

$$d_1 = 0,1d, d_2 = -0,02 - \text{ke gbe. puermer}$$

$$\text{Geben: } 0,18 \text{ m}$$

Demo: $\sim 9 (26.92)$

$$R = 0,2 \text{ m}$$

$$r = 0,1 \text{ m}$$

T-?

$$1) S_R = \pi R^2$$

$$S_r = \pi r^2$$

$$\frac{m}{M} = \frac{S_r}{S_R} \Rightarrow$$

$$\Rightarrow m = M \left(\frac{r}{R} \right)^2 = \frac{M}{4}$$

$$M' = M - \frac{M}{4} = \frac{3M}{4}$$

$$2) O_1 O_2 = 0,1 \text{ m}$$

$$d_{\text{ylong}} = \frac{M \cdot 0 - 0,1m}{M - m} = \left| - \frac{\frac{M}{4} \cdot 0,1}{\frac{3M}{4}} \right| = \left| - \frac{0,1}{3} \right| \approx 0,03 \text{ m}$$

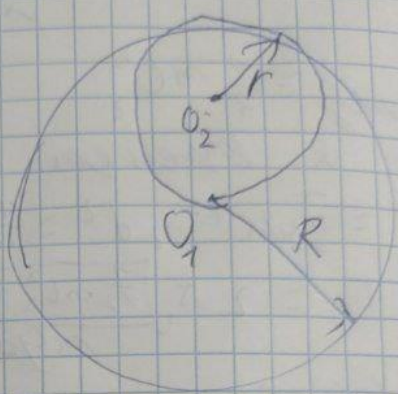
$$3) J_R = \frac{1}{2} M R^2$$

$$J_r = \frac{1}{2} m r^2 + m (R - r)^2 = \frac{1}{2} \frac{M}{4} 0,01 + \frac{M}{4} 0,07 =$$

$$= \frac{3M \cdot 0,01}{2 \cdot 4} \approx 0,00325 M$$

$$J' = J_R - J_r = \frac{1}{2} M R^2 - 0,00325 M =$$

$$\approx 0,01625 M$$



$$4) L = \frac{J'}{M'gd} = \frac{0,01625 \text{ M}}{\frac{3 \text{ M}}{4} \cdot 9,8 \cdot 0,3} \approx 0,065 \text{ m}$$

$$T = 2\pi \sqrt{\frac{L}{g}} = 2\pi \sqrt{\frac{0,065}{9,8}} \approx 0,511 \text{ s}$$

Antwort: 0,511 s